

Arrays 27

20 May 2026 13:13

```
class Solution {
    public int[] searchRange(int[] nums, int target) {
        int n = nums.length;
        int s = 0, e = n - 1;
        int low = -1;
        while (s <= e) {
            int mid = s + (e - s) / 2;
            if (nums[mid] == target) {
                low = mid;
                e = mid - 1;
            } else if (nums[mid] < target)
                s = mid + 1;
            else
                e = mid - 1;
        }
        s = 0;
        e = n - 1;
        int high = -1;
        while (s <= e) {
            int mid = s + (e - s) / 2;
            if (nums[mid] == target) {
                high = mid;
                s = mid + 1;
            } else if (nums[mid] < target)
                s = mid + 1;
            else
                e = mid - 1;
        }
        return new int[] { low, high };
    }
}
```

Koko eating banana

```
class Solution {
    public boolean canEat(int[] piles, int h, int k) {
        long total_hours = 0;
        for (int i = 0; i < piles.length; i++) {
            total_hours += piles[i] / k;
            if (piles[i] % k != 0) total_hours++;
        }
        if (total_hours <= h) return true;
        return false;
    }
    public int minEatingSpeed(int[] piles, int h) {
        int n = piles.length;
        int maxi = Integer.MIN_VALUE;
        for (int i = 0; i < n; i++) {
            maxi = Math.max(maxi, piles[i]);
        }
        int s = 1, e = maxi;
        int ans = -1;
        while (s <= e) {
            int mid = s + (e - s) / 2;
            if (canEat(piles, h, mid) == true) {
                ans = mid;
                e = mid - 1;
            } else s = mid + 1;
        }
        return ans;
    }
}
```

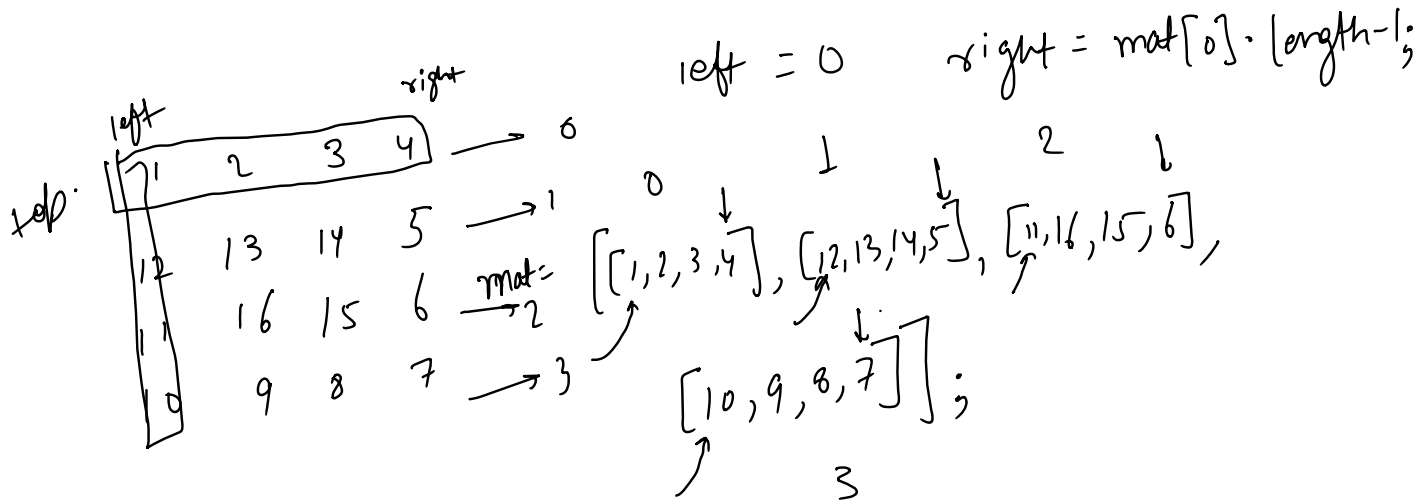
Spiral Matrix

```
public class Array_2d_second {
    public static void main(String[] args) {
        int [][] mat = {{1,2,3},{4,5,6},{7,8,9}};
        int left=0, right=mat[0].length-1;
        int top=0, bottom=mat.length-1;
        while (top <= bottom && left <= right) {
            for (int i = left; i <= right; i++) {
                System.out.print(mat[top][i]);
            }
        }
    }
}
```

```

    }
    top++;
    for(int i=top; i<=bottom; i++){
        System.out.print(mat[i][right]);
    }
    right--;
    if(top<=bottom){
        for(int i=right; i>=left; i--){
            System.out.print(mat[bottom][i]);
        }
    }
    bottom--;
    if(left<=right){
        for(int i=bottom; i>=top; i--){
            System.out.print(mat[i][left]);
        }
    }
    left++;
}
}
}

```



int top=0, bottom=mat.length-1;

for(int i=left; i<=right; i++){

 Sout(mat[top][i]);

 ↓ ↓

row col

top++;

for(int i=top; i<=bottom; i++){

 Sout(mat[i][right]);

✓

```
    cout ( mat [i][right] );  
    }  
    right --;
```